Three Research Ideas

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Idea 1: Anatomically Guided Semi-Supervised Learning for Retinal OCT Disease Classification

1. The Problem

Deep learning models achieve high accuracy in retinal Optical Coherence Tomography (OCT) classification but rely on large, expert-annotated datasets. Generating pixel-level or image-level annotations requires experienced ophthalmologists, making dataset creation expensive. While semi-supervised learning (SSL) leverages unlabeled images to reduce annotation needs, existing SSL methods primarily use confidence-based pseudo-labeling and consistency regularization. These approaches often ignore the anatomical structure of retinal layers, producing incorrect pseudo-labels when disease characteristics are subtle or visually similar, which hinders reliable clinical deployment.

2. The Research Gap

Recent work like SD-RetinaNet (2025) demonstrates the value of topological and anatomical constraints for robust retinal biomarker segmentation.[1] However, a key limitation is that these rigid anatomical priors are primarily designed for specific segmentation tasks and do not easily generalize to multi-class disease classification. The authors note that extending such constraints to classification is challenging, as overlapping visual features cause confidence-based pseudo-labeling to fail when the broader anatomical context is ignored. Thus, a clear gap exists in developing adaptive, anatomy-aware SSL specifically for OCT classification.

3. Proposed Solution

I propose an Anatomically Guided SSL framework combining attention-based feature extraction with anatomy-aware consistency regularization. A lightweight attention network will first identify clinically important retinal regions, weighting pseudo-label confidence according to anatomical relevance rather than global prediction scores. The model will incorporate adaptive confidence thresholds and uncertainty estimation to filter unreliable pseudo-labels dynamically. By explicitly integrating retinal layer topology into the pseudo-labeling process, this approach aims to improve label efficiency, increase robustness under limited supervision, and provide clinically reliable retinal disease classification.

Idea 2: Adaptive Federated Semi-Supervised Learning for Privacy-Preserving Medical Image Analysis

1. The Problem

Medical institutions possess vast imaging datasets, but strict privacy regulations prevent centralized data sharing. Federated Learning (FL) enables collaborative training without exchanging raw data, yet current FL systems still require substantial labeled datasets at each hospital. Many hospitals only have a small amount of annotated data alongside large unlabeled archives. Consequently, existing federated semi-supervised learning (FSSL) methods fail to fully utilize available medical data and often suffer significant performance degradation under heterogeneous, non-IID (non-independent and identically distributed) data distributions across different clinical sites.

[1] `https://ieeexplore.ieee.org`

2. The Research Gap

Recent studies, such as Wang et al.'s work on Robust Federated Semi-Supervised Learning (2024), propose pseudo-label filtering to improve FSSL robustness.[2] However, a major limitation identified in this domain is that current filtering strategies apply static or uniform confidence thresholds across all participating clients. This approach fails to dynamically adapt to severe non-IID data distributions and varying local annotation qualities. The literature suggests that developing adaptive, client-specific learning strategies that improve pseudo-label reliability while preserving privacy remains a critical unresolved challenge.

3. Proposed Solution

I propose an Adaptive FSSL framework where each client dynamically adjusts pseudo-label confidence thresholds based on local data quality and model uncertainty. Instead of applying global thresholds, the server will aggregate only high-confidence, client-specific representations, reducing noise propagation across heterogeneous nodes. The framework will also integrate privacy-preserving techniques like differential privacy or secure aggregation to ensure compliance with healthcare regulations. This research aims to improve diagnostic performance across diverse hospital networks while maintaining patient privacy and drastically reducing annotation costs.

Idea 3: Lightweight Explainable Hybrid Vision Transformers for Medical Image Classification

1. The Problem

Vision Transformers (ViTs) demonstrate impressive performance in medical image analysis, but most architectures are computationally expensive and difficult to deploy in hospitals with limited hardware resources. Furthermore, transformer-based models often lack interpretable decision-making processes, making clinicians hesitant to trust automated predictions. Balancing computational efficiency, high diagnostic accuracy, and clinical explainability remains a significant challenge, as current models force a trade-off between the local feature extraction of CNNs and the global context modeling of transformers.

2. The Research Gap

Recent hybrid models like HyCoViT (2025) successfully combine convolutional networks and Vision Transformers for enhanced medical image classification.[3] However, the authors explicitly note a limitation regarding interpretability, stating that while hybrid models improve accuracy, they struggle to provide clinically meaningful, high-resolution explainability maps without introducing significant computational overhead.[4] Many recent papers highlight that improving visual interpretability for hybrid architectures without sacrificing the computational efficiency required for edge deployment remains an open and critical research direction.

3. Proposed Solution

I propose a highly efficient hybrid architecture that merges lightweight CNN backbones for local anatomical feature extraction with efficient transformer modules for long-range dependency modeling. Crucially, I will integrate a specialized explainability module utilizing gradient-weighted attention visualization to highlight specific regions influencing predictions, enabling clinicians to verify decisions transparently. To ensure deployment feasibility, I will apply structured pruning and knowledge distillation to compress the model without losing diagnostic accuracy. This framework will deliver accurate, interpretable, and resource-efficient medical image classification suitable for real clinical settings.

[2] https://www.researchgate.net

[3] https://www.cse.scu.edu

[4] https://www.researchgate.net